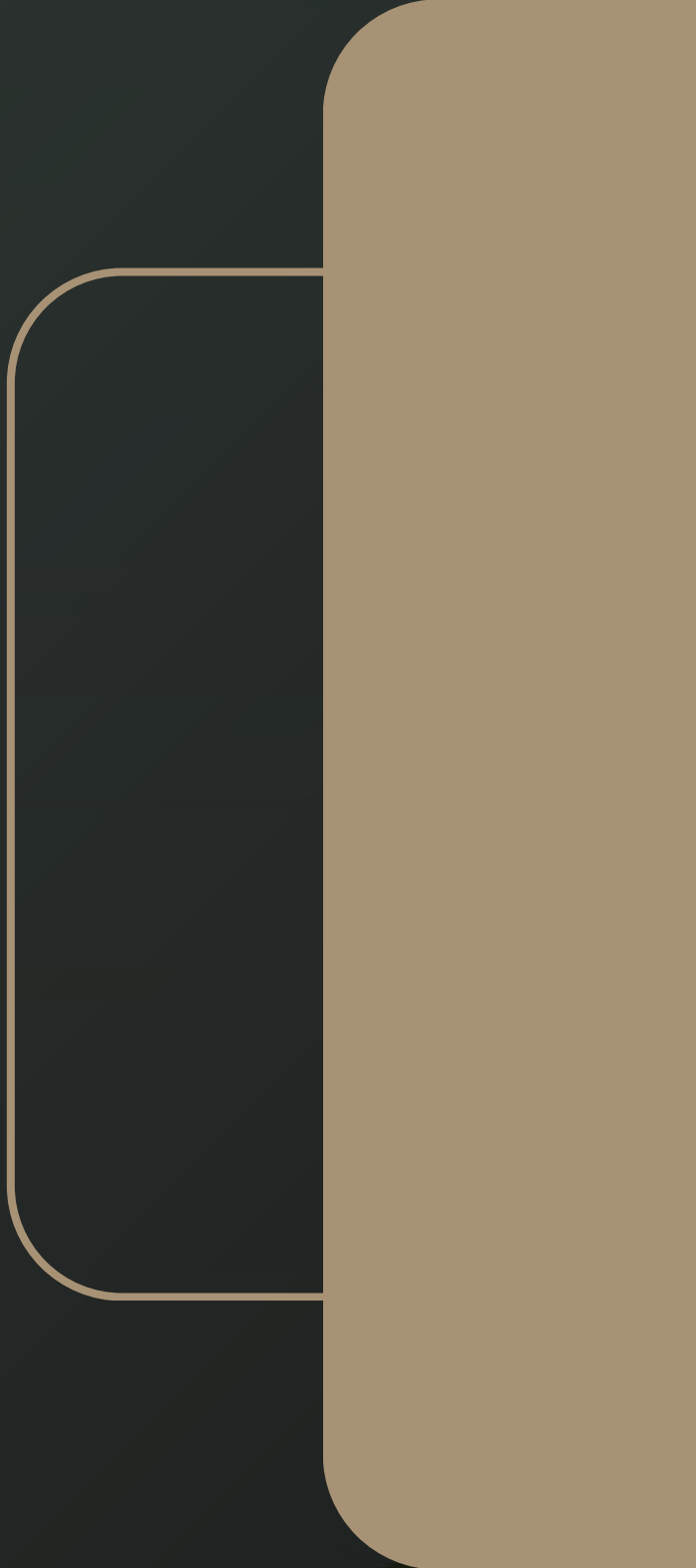


UML DIAGRAMS

USE CASE & ACTIVITY DIAGRAM



Yanitsa Ivanova





AGENDA



01.

Intro & History

02.

Use Case Diagrams

03.

Activity Diagrams

04.

Capstone &
Summary

WHAT IS UML?

Unified Modeling Language (UML) is a standardized visual language for modeling software systems.

- Visualize software systems
- Specify requirements and behavior
- Design before implementation
- Document systems
- Communicate ideas across teams

"UML is a language for specifying, visualizing, constructing, and documenting the artifacts of software systems."

UML is a modeling language, not a programming language.





HISTORY OF UML:

THE PROBLEM

- 1980s–1990s: Dozens of competing modeling methods emerged
- By 1994, over 50 different methods
- The "Method Wars": No standard language existed, making collaboration and communication across teams and projects extremely difficult.
- Three dominant methods stood out before unification

HISTORY:

THE THREE AMIGOS

Grady Booch — Booch

Method

- Object-oriented design
- Detailed system design
- Implementation
- Engineering-intensive applications

James Rumbaugh — OMT

- Object Modeling Technique
- System analysis
- Data-intensive applications
- Modeling relationships between objects

Ivar Jacobson — OOSE

- Object-Oriented SE
- Requirements analysis
- Use cases
- User interactions
- Describing system behavior from the user's perspective

UML unified useful ideas from multiple object-oriented methods into a common modeling language.

USE CASE DIAGRAMS



A Use Case Diagram gives a high-level view of system functionality.

WHO?

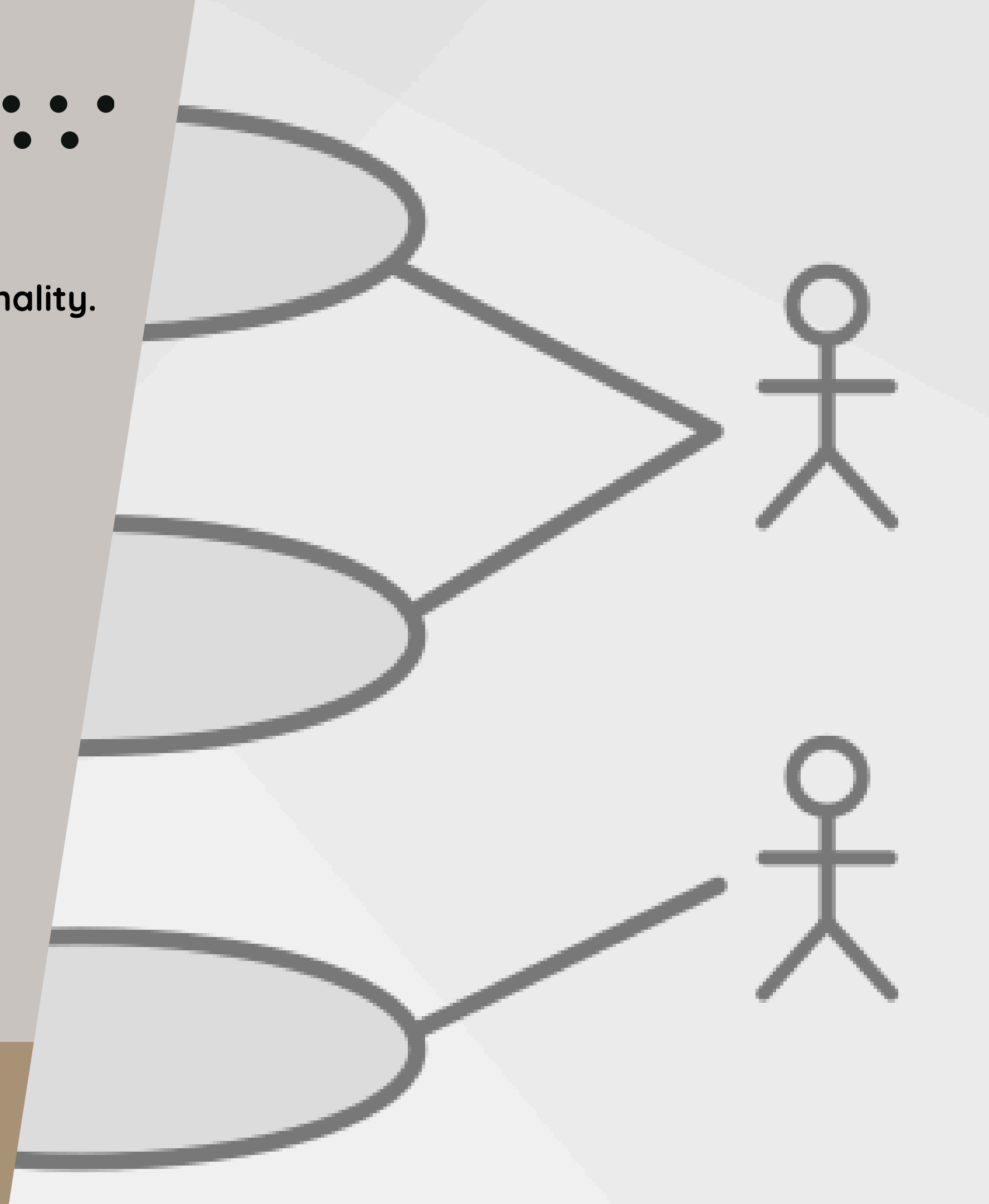
Who interacts with the system?

WHAT?

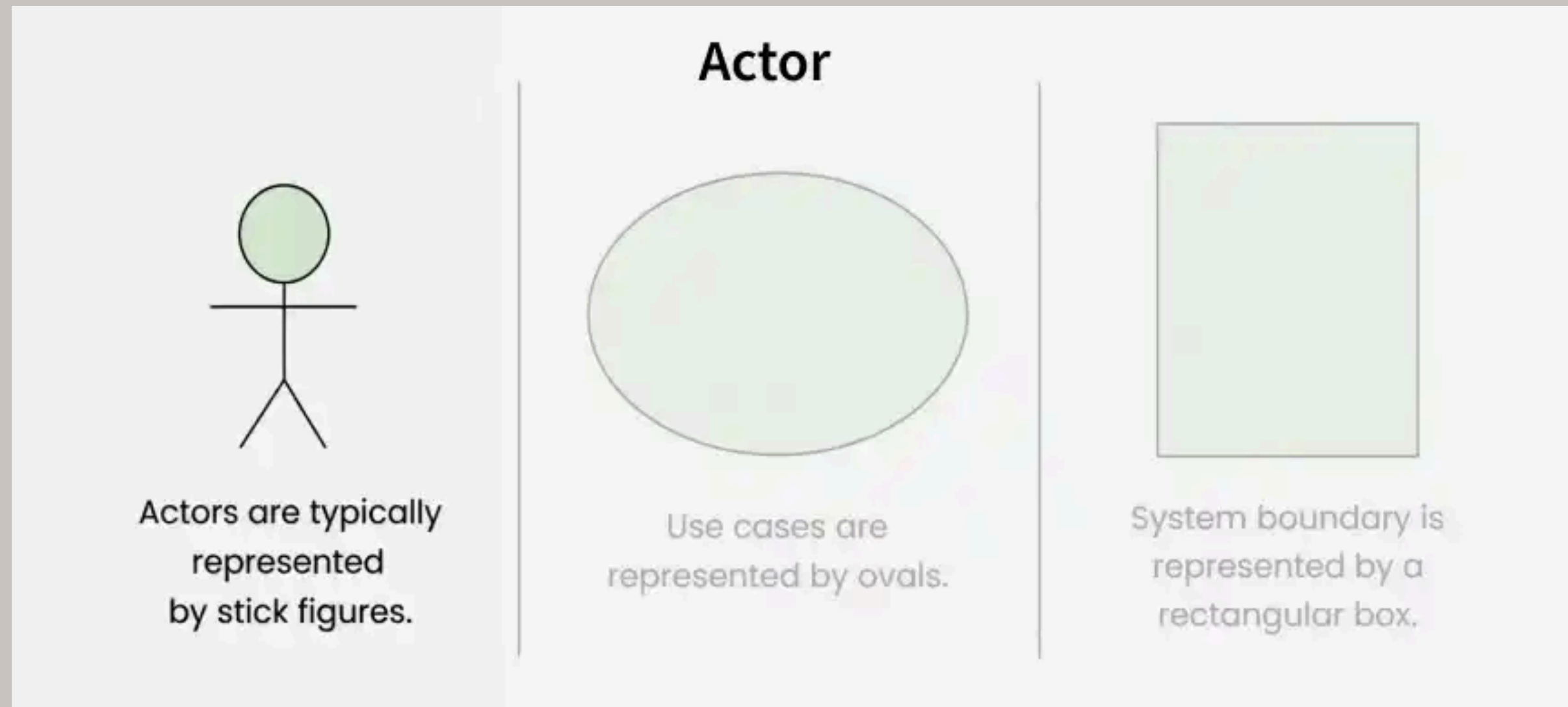
What can they do?

Common Uses

- Identify system requirements
- Identify users/roles
- Define system boundaries
- Communicate functionality to stakeholders
- Establish the scope of a project



USE CASE DIAGRAM: NOTATIONS



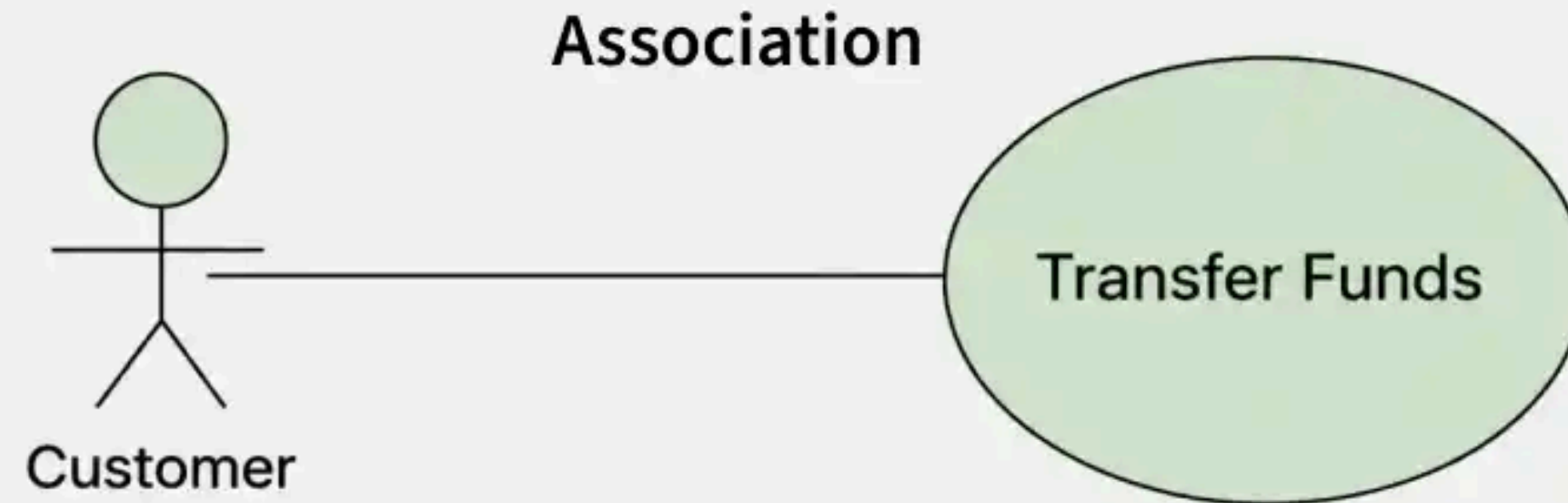
USE CASE DIAGRAM: NOTATIONS



USE CASE DIAGRAM: NOTATIONS



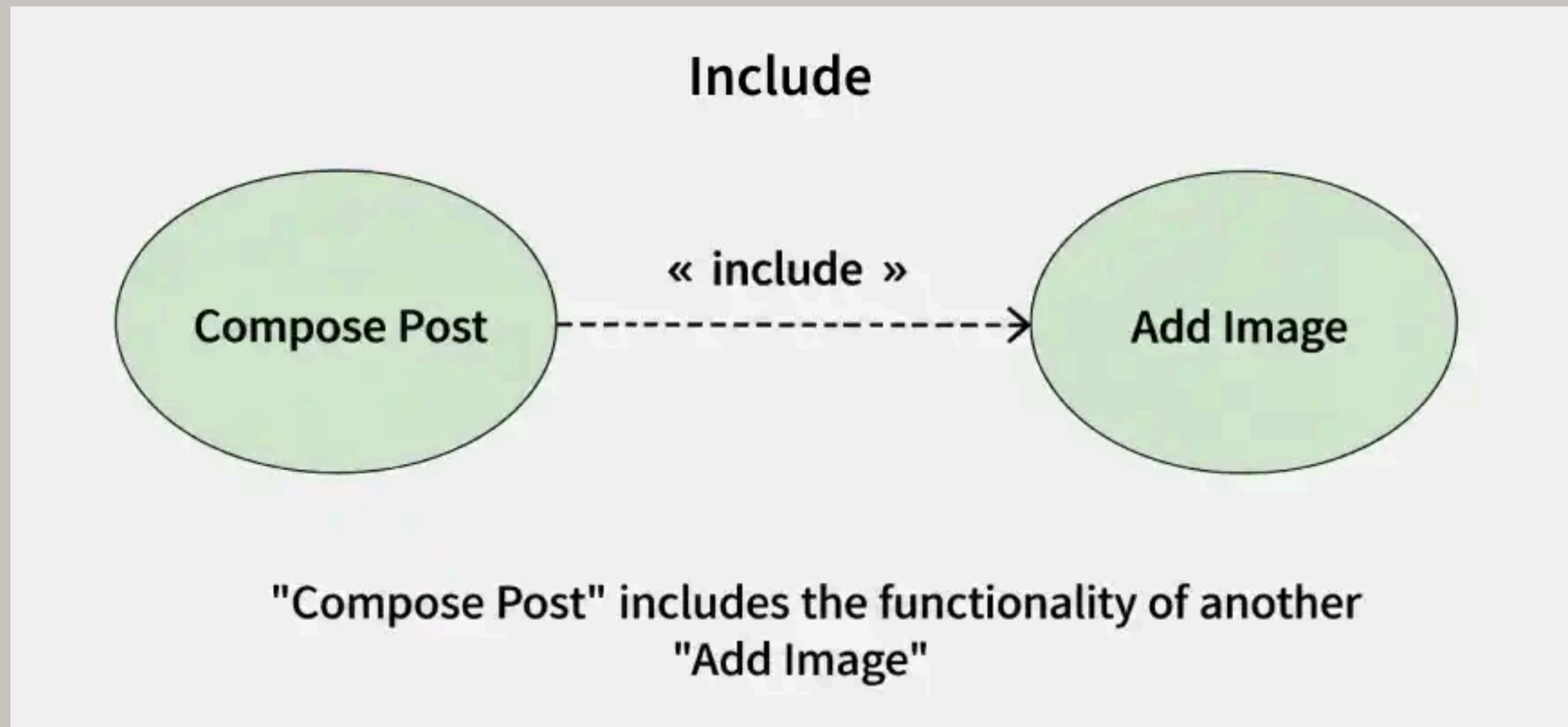
USE CASE DIAGRAM: RELATIONSHIPS



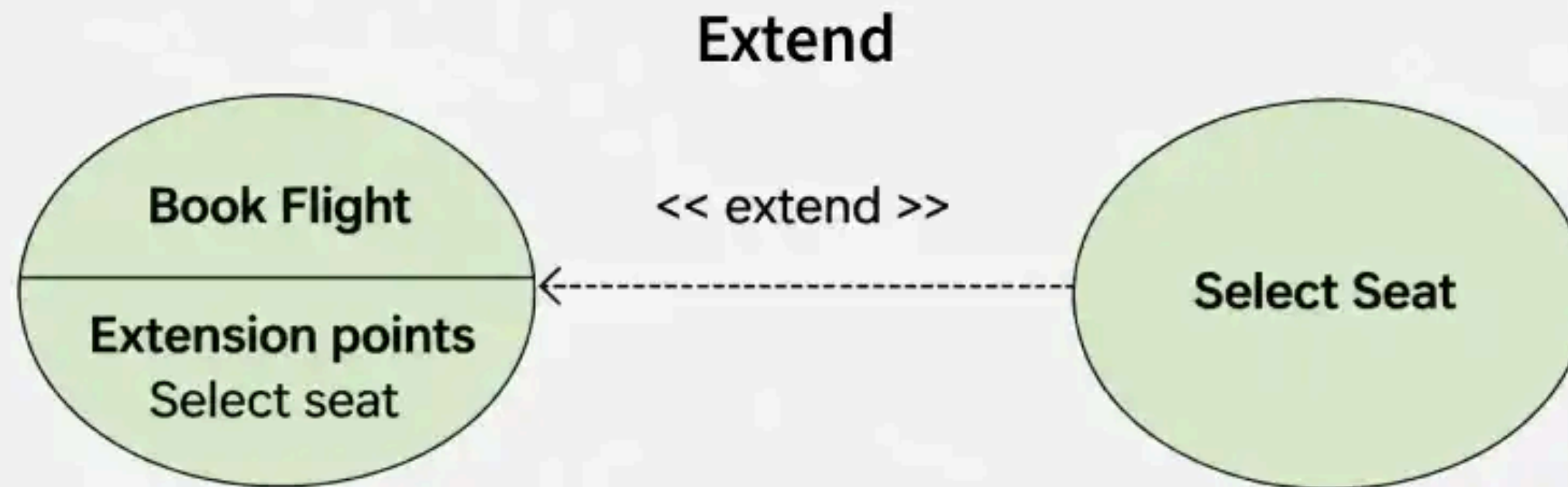
Represents a communication or interaction between an actor(Customer) and a use case(Transfer Funds)



USE CASE DIAGRAM: RELATIONSHIPS



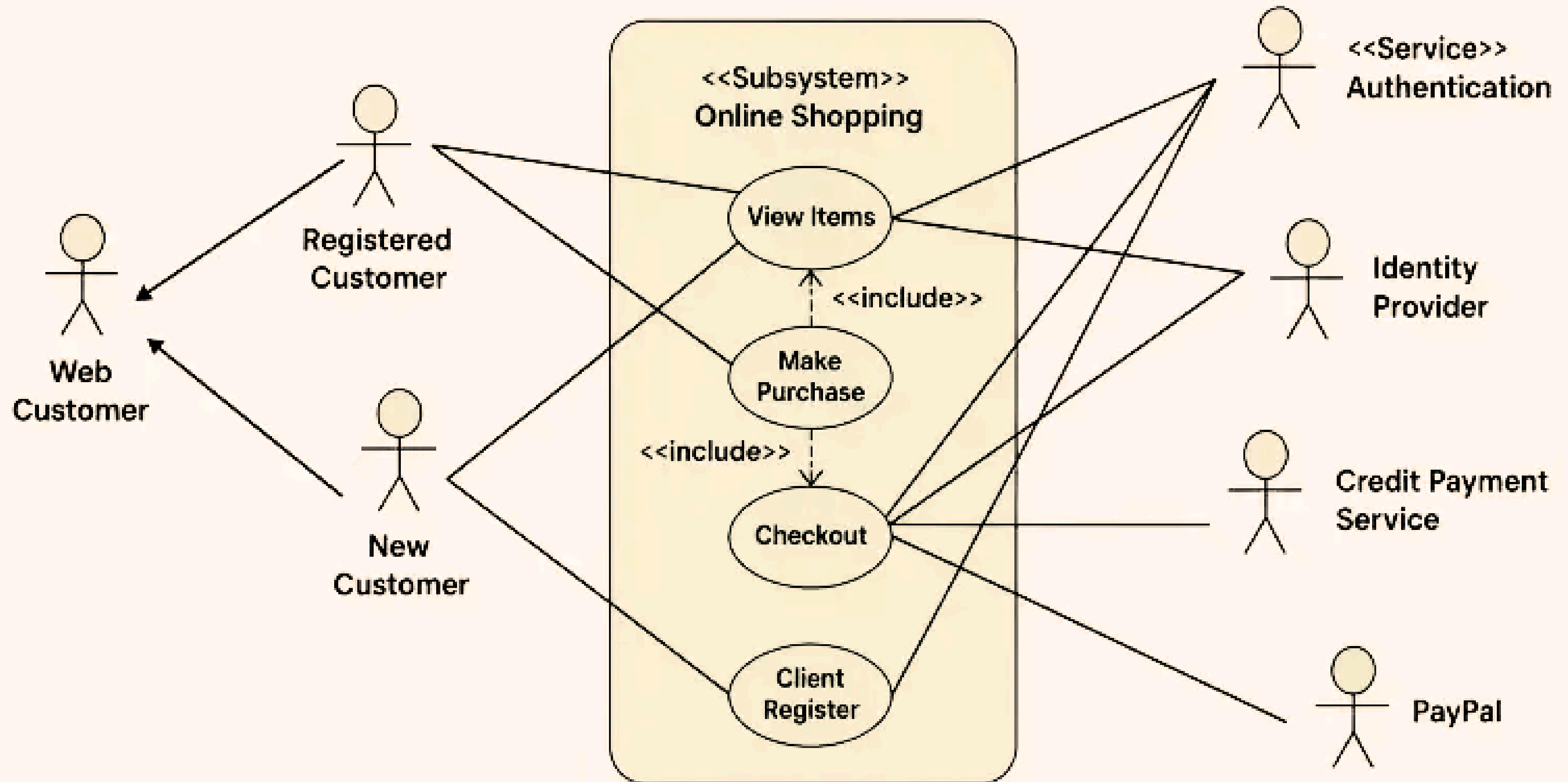
USE CASE DIAGRAM: RELATIONSHIPS



The “Select Seat” use case may extend the “Book Flight” use case when the user wants to choose a specific seat.



Use Case diagram of an Online Shopping System



ACTIVITY DIAGRAM

An Activity Diagram shows the workflow or sequence of activities within a process.

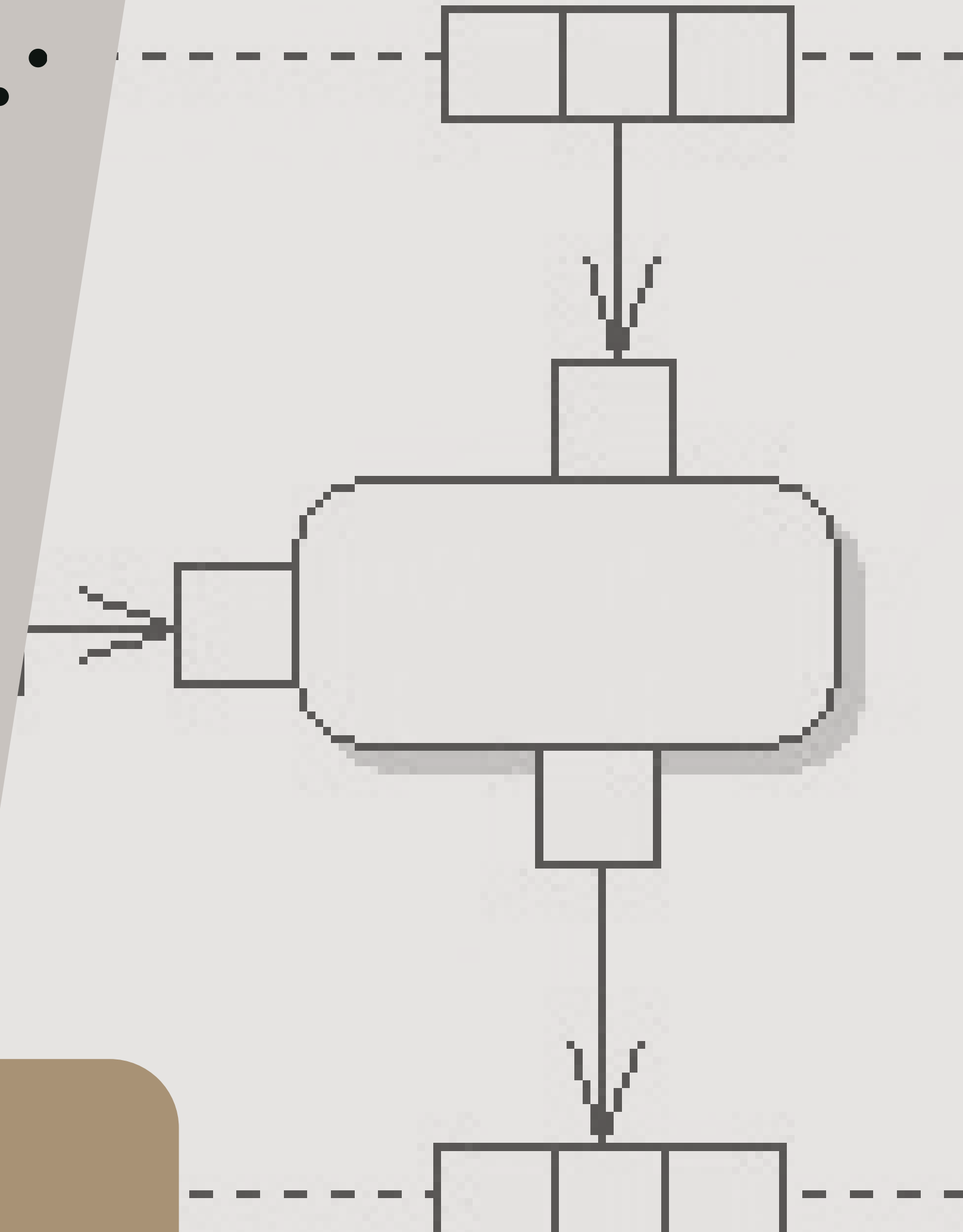
Use Case Diagrams ask
“What can the system do?”

Activity Diagrams ask
“How does this process
happen?”

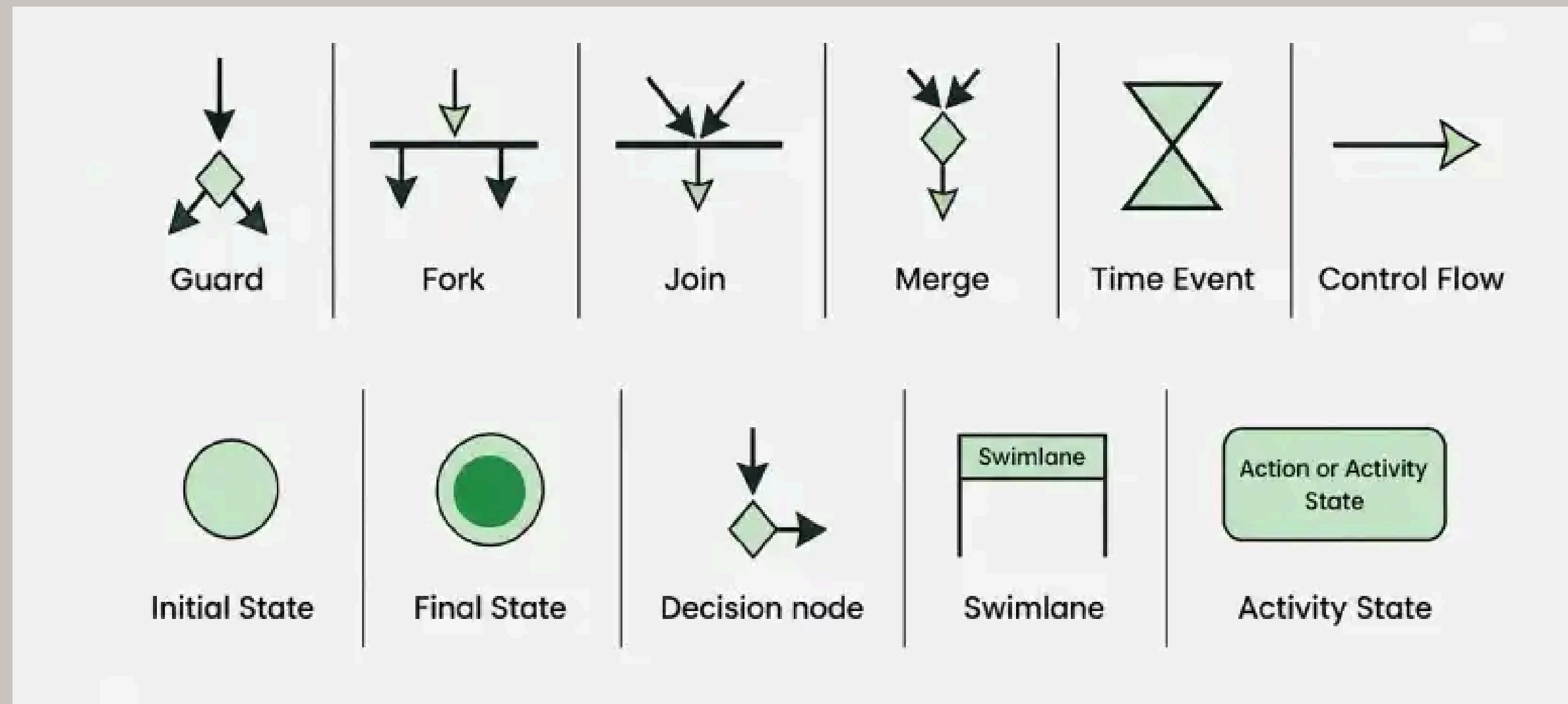
Common Uses

- Model business processes
- Model user workflows
- Show system behavior
- Identify decisions
- Show alternative paths
- Document complex processes

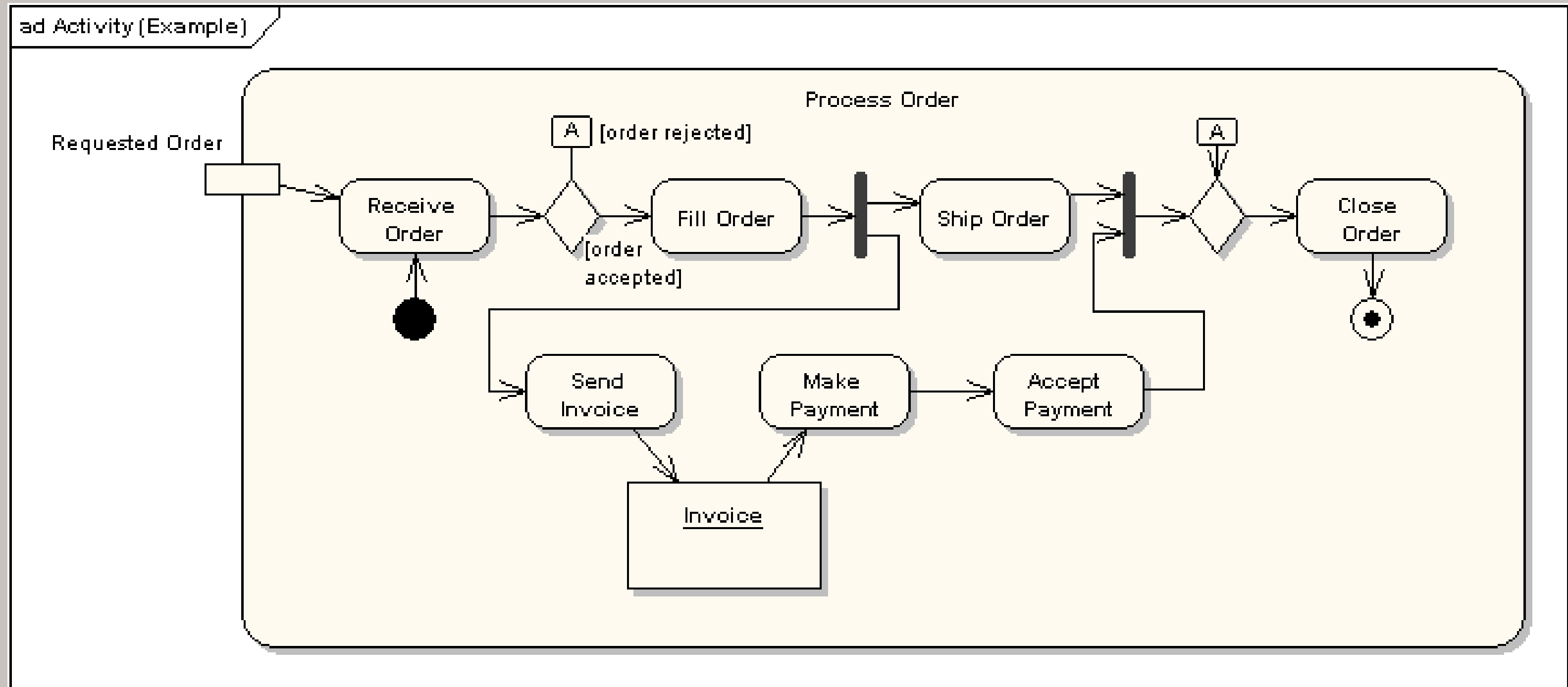
UML activity diagrams represent step-by-step workflows and the overall flow of control.



ACTIVITY DIAGRAM: NOTATIONS



ACTIVITY DIAGRAM: EXAMPLE



UML IN CAPSTONE PROJECTS

Use Case Diagram

HIGH-LEVEL OVERVIEW OF THE
SYSTEM

WHAT FEATURES EACH USER CAN
ACCESS

WHO WILL USE THE SYSTEM

IDENTIFY THE MAIN
REQUIREMENTS

Activity Diagram

HOW A PROCESS WORKS STEP BY
STEP

SHOW DECISIONS OR POSSIBLE
OUTCOMES IN A PROCESS

WHAT HAPPENS AFTER A USER
COMPLETES AN ACTION

PLAN THE WORKFLOW BEHIND A
FEATURE

SUMMARY

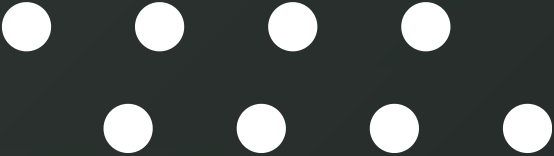
Use Case Diagram

- Shows what the system can do
- Focuses on users/actors and their goals
- Gives a high-level overview

Activity Diagram

- Shows how a process works step by step
- Focuses on actions, decisions, and flow
- Gives a detailed workflow

"UML lets us visualize the system before we have to explain it in code."



REFERENCES

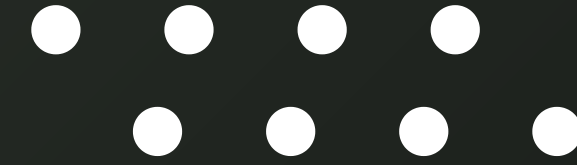
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THANK YOU

